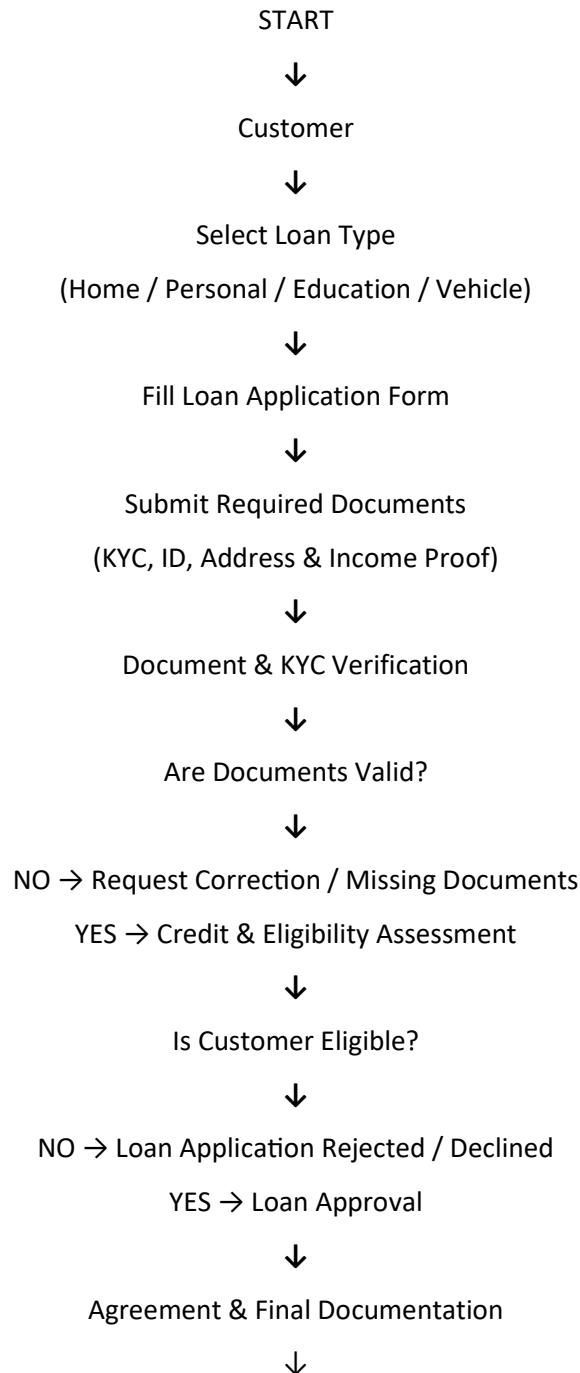


Conversational AI Assignment - 1

1. Draw and explain process flow diagram for a customer who is applying loan from a bank.

Sol- Process Flow Diagram of Customer Applying for a Bank Loan

The following flow represents the typical process followed by a customer when applying for a loan from a bank.



Loan Disbursement



END

Explanation of the Loan Application Process

1. Loan Selection:

The customer first selects the type of loan they want to apply for, such as a personal loan, home loan, education loan, or vehicle loan. The customer chooses the loan based on their financial requirement. Each type of loan may have different eligibility criteria, interest rates, repayment periods, and documentation requirements.

2. Application:

After selecting the loan type, the customer fills out the loan application form. The customer provides important details such as their name, contact information, address, employment details, monthly income, required loan amount, and other personal and financial information. This information helps the bank understand the customer's financial situation.

3. Document Submission:

The customer submits the documents required by the bank to support the information provided in the application. These may include identity proof, address proof, income proof, bank statements, employment documents, and other relevant documents. The required documents may vary depending on the type and amount of the loan.

4. KYC & Document Verification:

The bank checks and verifies the customer's submitted documents and information. KYC (Know Your Customer) verification is performed to confirm the customer's identity and address. The bank also checks whether the documents are valid, complete, and consistent with the information provided in the application. If any document is missing or incorrect, the customer may be asked to provide the correct information or submit the missing document.

5. Eligibility Assessment:

Once the documents are successfully verified, the bank evaluates whether the customer is eligible for the loan. The bank may consider factors such as the customer's income, employment status, credit history or credit score, existing loans or liabilities, repayment capacity, and requested loan amount. Based on these factors, the bank determines whether the customer can reasonably repay the loan.

6. Approval or Rejection:

After completing the eligibility and credit assessment, the bank makes a decision on the loan application. If the customer satisfies the required eligibility and credit criteria, the loan is approved. If the customer does not meet the required criteria, the application may be rejected or declined. The bank then communicates the decision to the customer.

7. Agreement:

If the loan is approved, the customer reviews and accepts the loan terms and conditions provided by the bank. These may include the approved loan amount, interest rate, repayment period, fees, and repayment schedule. The customer then signs the required loan agreement and completes any remaining documentation before the loan can be disbursed.

8. Loan Disbursement:

After all the required formalities and documentation are completed, the bank proceeds with loan disbursement. The approved loan amount is transferred according to the terms of the loan. Depending on the type of loan, the amount may be credited to the customer's bank account or transferred directly to the relevant seller, institution, or other party. This completes the loan application process.

2. Propose a Conversational AI solution for a hospital to reduce receptionist Workload.

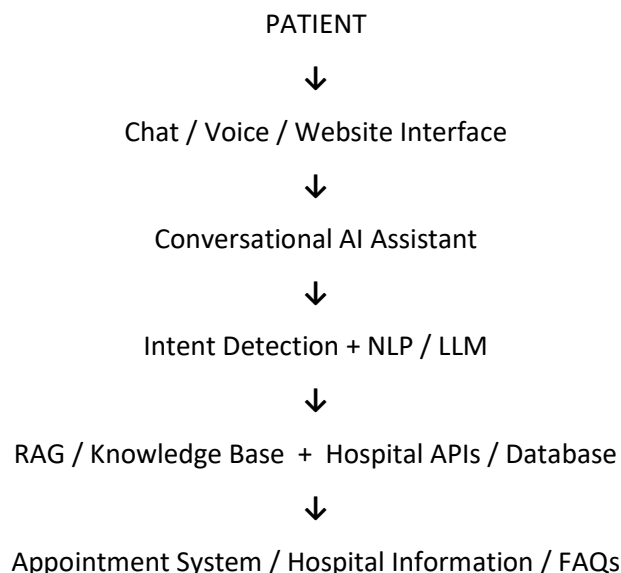
Problem Statement

Hospitals receive a large number of calls and queries every day regarding appointments, doctor availability, cancellations, rescheduling, hospital timings, departments, location, and other frequently asked questions. Handling these repetitive queries manually increases the workload of receptionists and may increase patient waiting time.

Proposed Solution: Hospital Conversational AI Assistant

An AI-powered hospital chatbot or voice assistant can act as the first point of contact for patients. It can understand natural-language queries, retrieve information, perform routine actions through hospital systems, and transfer complex cases to a human receptionist.

High-Level Architecture





Response to Patient

How the System Works:-

Example 1 – Appointment Booking

Patient: "I want to book an appointment with a cardiologist."

AI Assistant: "Sure. Which date would you prefer?"

The system checks doctor availability and provides available time slots. The patient selects a slot and the AI confirms the appointment.

Example 2 – Rescheduling

Patient: "I cannot attend my appointment tomorrow. Can I reschedule it?"

The AI verifies the appointment and provides available alternative dates and time slots.

Example 3 – General Query

Patient: "What time does the hospital open?"

AI Assistant: "The hospital OPD is open from 9 AM to 5 PM, Monday to Saturday."

Key Features

Feature	Purpose
Appointment Booking	Automatically book appointments.
Cancellation	Cancel existing appointments.
Rescheduling	Change appointment dates or times.
Doctor Search	Find doctors by department or specialization.
Availability	Show available appointment slots.
Hospital FAQs	Answer common hospital-related questions.
Location & Directions	Provide hospital and department location information.
Notifications	Send appointment reminders and confirmations.
Patient Verification	Verify patients before accessing personal information.
Human Handoff	Transfer complex queries to a receptionist or appropriate staff member.

AI Components

LLM: Understands natural-language questions and generates conversational responses.

RAG: Retrieves accurate information from hospital FAQs, departments, policies, and services.

Backend APIs: Handles actions such as appointment booking, cancellation, and rescheduling.

Database: Stores appointment and other authorized patient information.

Authentication: Protects patient-specific and sensitive information.

Human Handoff: Escalates complex or sensitive queries to hospital staff.

Benefits

- Reduces repetitive workload for receptionists.
- Provides assistance 24×7 for routine queries.
- Reduces patient waiting time.
- Automates appointment booking, cancellation, and rescheduling.
- Provides consistent responses to common questions.
- Handles multiple patient conversations simultaneously.
- Allows receptionists to focus on complex patient requirements.

Conclusion

The proposed Conversational AI system acts as a virtual hospital receptionist. It automates repetitive tasks such as appointment booking, cancellation, rescheduling, doctor availability, and FAQ handling. By integrating an LLM/RAG system with hospital APIs and databases, the solution can provide fast and personalized assistance while allowing human receptionists to focus on more complex patient requirements.